# Distributivity and the Commensurability of Value-Definite Realism and Empirical Adequacy

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**Abstract**

When does an empirically adequate theory admit a value-definite completion — an assignment of simultaneous definite values to all observables? The answer depends on the algebraic structure of the observations. For a directed system of Boolean algebras carrying compatible charges, the Stone construction produces an unconditional measure on the dual space, and every observable admits a simultaneous definite value via ultrafilters. The passage to a probability measure on an underlying state space requires only $\sigma$-additivity, and the choice of state space is unconstrained by the algebra [12].

For orthomodular lattices — the algebraic framework for incompatible observations — the situation is structurally different. The McDonald–Bimbó dual space carries principal filters as distinguished structure, and the Kochen–Specker theorem blocks global value-definiteness. Probabilistic realism (descent to a probability measure) remains available via Gleason-type results, but value-definite realism does not.

The dividing line is distributivity. The state space admits a natural filtration $\mathcal{S} \supseteq \mathcal{S}_\sigma \supseteq \mathcal{S}_{\mathrm{df}}$, whose three transitions — pasting, regularity, sharpness — are of different mathematical character. Distributivity is the exact condition under which the filtration degenerates: all three obstructions vanish, and value-definite realism becomes commensurable with empirical adequacy.

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## 1 Introduction

A central question in the philosophy of science asks when an empirically successful theory can be completed to a realist one — one that assigns definite properties to an underlying reality, not merely adequate predictions to observable phenomena. Van Fraassen [15] argues that science need only aim at *empirical adequacy*: agreement with all observable phenomena. The scientific realist demands more — an underlying state space whose properties generate the observations.

This debate is usually conducted in philosophical terms. We give it a mathematical formulation.

Three nested positions, each a strengthening of the previous:

(i) **Empirical adequacy (EA).** A measure on the dual space of the observation algebra that agrees with all observed charges. No commitment to an underlying state space.

(ii) **Probabilistic realism (PR).** A probability measure on an underlying state space $(\Omega, \sigma(\mathcal{C}))$ that generates the observations. Requires descent from the dual-space measure.

(iii) **Value-definite realism (VDR).** Every observable simultaneously has a definite value. A global dispersion-free state — equivalently, a 2-valued homomorphism on the full observation algebra.

These are nested: VDR implies PR implies EA. Which implications reverse depends on the observation algebra:

*Compatible observations.* When all observations can be performed simultaneously, the observation algebra is a Boolean algebra — a complemented distributive lattice. Every Boolean algebra admits 2-valued homomorphisms (ultrafilters exist by Zorn's lemma). In this setting, all three positions are available and mutually commensurable. The passage from EA to PR requires $\sigma$-additivity of the charges [12]; the further passage to VDR is free.

*Incompatible observations.* When some observations cannot be performed simultaneously — as in quantum mechanics, contextual experimental design, or distributed systems with interfering sensors — the observation algebra is an orthomodular lattice (OML), a complemented but non-distributive lattice. The Kochen–Specker theorem [11] shows that for the prototypical case $L(H)$ with $\dim H \geq 3$, no 2-valued homomorphism exists: VDR is blocked. Probabilistic realism remains available via Gleason's theorem [6], and empirical adequacy is unconditional.

The dividing line is distributivity. The individual ingredients — Stone duality, the Kochen–Specker theorem, Gleason's theorem, the Bub–Clifton uniqueness result, and the recent McDonald–Bimbó duality for orthomodular lattices [13] — are known. What has not been stated is the synthesis: that distributivity of the observation algebra is the condition controlling which of the three positions are available, and that the state space admits a filtration whose transitions are of different mathematical character. We begin with the Boolean case [12].

Let $(\iota, \leq)$ be a directed set indexing a family of Boolean algebras $\{B_i\}_{i\in\iota}$ with injective homomorphisms $\varphi_{ij} : B_i \to B_j$ for $i \leq j$. Write $\mathcal{C} = \bigcup_i B_i$ for the direct limit — the *cylinder algebra.*

A *compatible family of normalised charges* is a collection $\{\ell_i\}$ with each $\ell_i : B_i \to [0, 1]$ finitely additive, $\ell_i(1) = 1$, and

$$\ell_j \circ \varphi_{ij} = \ell_i \qquad (i \leq j).$$

The Stone space $\mathrm{St}(\mathcal{C})$ is the space of ultrafilters on $\mathcal{C}$, compact and totally disconnected. Each $\omega \in \Omega$ determines a principal ultrafilter

$$\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}.$$

Any compatible family of normalised charges extends to a $\sigma$-additive Baire probability measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ — unconditionally, with no assumption of $\sigma$-additivity [3, Theorem 1], [12]. This is the *empirically adequate* object: it agrees with all charges on cylinder events.

The measure $\hat{\mu}$ lives on $\mathrm{St}(\mathcal{C})$, not on $\Omega$. Descent to a probability measure $P$ on $(\Omega, \sigma(\mathcal{C}))$ requires

$$\hat{\mu}\big(\mathrm{pure}(\Omega)\big) = 1,$$

i.e., that $\hat{\mu}$ concentrates on the principal ultrafilters. This holds if and only if each $\ell_i$ is $\sigma$-additive [12].

Every Boolean algebra admits ultrafilters (by Zorn's lemma), and every ultrafilter is a 2-valued homomorphism. A 2-valued homomorphism assigns to each element $a \in \mathcal{C}$ a definite value in $\{0, 1\}$, respecting the lattice operations.

Moreover, any subset $S \subseteq \mathrm{St}(\mathcal{C})$ that projects surjectively onto each $\mathrm{St}(B_i)$ can serve as a realisation space: set $\Omega = S$ with evaluation maps given by the restriction maps. The algebra imposes no constraint on the choice of $\Omega$ [12].

**Proposition 1.1** (Boolean commensurability)**.** *For a directed system of Boolean algebras with compatible charges:*

*(a) EA is unconditional: $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ always exists.*

*(b) PR holds iff each $\ell_i$ is $\sigma$-additive.*

(c) *VDR holds unconditionally:* $\mathrm{St}(\mathcal{C})$ *consists entirely of 2-valued homomorphisms.*

*The only obstruction is $\sigma$-additivity at the EA–PR transition, and it is a condition on the charges, not the algebra.*

*Proof.* Part (a) is the Stone construction [3, 12]. Part (b) is [12, Theorem 1]. Part (c): ultrafilters on a Boolean algebra are exactly the 2-valued homomorphisms; their existence follows from Zorn's lemma applied to the filter generated by any non-zero element. □

## 2 Orthomodular observation algebras

Proposition 1.1 depends on distributivity at exactly one point: the existence of 2-valued homomorphisms. When observations are incompatible, this fails.

**Definition 2.1.** An *ortholattice* is a bounded lattice $A = (A, \wedge, \vee, {}^{\perp}, 0, 1)$ with an orthocomplementation ${}^{\perp} : A \to A$ satisfying $a \wedge a^{\perp} = 0$, $a \vee a^{\perp} = 1$, $a^{\perp\perp} = a$, and $a \leq b \Rightarrow b^{\perp} \leq a^{\perp}$.

**Definition 2.2.** An *orthomodular lattice* (OML) is an ortholattice satisfying the *orthomodular law*:

$$a \leq b \quad \Longrightarrow \quad b = a \vee (a^{\perp} \wedge b).$$

Every Boolean algebra is an OML (the orthomodular law follows from distributivity), but the converse fails. An OML is Boolean if and only if it is distributive [13, Proposition 2.3].

**Example 2.3.** The lattice $L(H)$ of closed linear subspaces of a separable Hilbert space $H$ with $\dim H \geq 3$, ordered by inclusion with $U^{\perp} = \{x \in H : \langle x, y\rangle = 0 \ \forall y \in U\}$, is an orthomodular lattice that is not distributive [10, Chapter 1].

**Example 2.4** (Incompatible measurements)**.** Three binary measurements $A$, $B$, $C$, where $(A, B)$ and $(A, C)$ can be performed jointly but $(B, C)$ cannot. The jointly measurable pairs generate Boolean subalgebras; the full observation algebra is orthomodular, not Boolean.

Two elements $a, b$ in an OML *commute* if they generate a Boolean subalgebra. A maximal set of pairwise commuting elements forms a *Boolean context* — a maximal Boolean subalgebra of $A$.

**Definition 2.5.** A *state* on an OML $A$ is a function $s : A \to [0, 1]$ with $s(1) = 1$ and

$$a \perp b \quad \Longrightarrow \quad s(a \vee b) = s(a) + s(b),$$

where $a \perp b$ means $a \leq b^{\perp}$.

A state is additive on *orthogonal* pairs only — not on arbitrary joins. This is the OML analogue of a finitely additive charge on a Boolean algebra.

**Definition 2.6.** A state $s$ is *dispersion-free* if $s(a) \in \{0, 1\}$ for all $a \in A$.

A dispersion-free state is a 2-valued homomorphism: it assigns a definite truth value to every proposition in $A$. On a Boolean algebra, dispersion-free states are exactly the ultrafilters.

**Theorem 2.7** (Kochen–Specker [11])**.** *If* $\dim H \geq 3$*, there is no dispersion-free state on* $L(H)$*.*

For $\dim H = 2$, dispersion-free states exist; the obstruction is specific to $\dim H \geq 3$.

**Theorem 2.8** (Gleason [6])**.** *If* $\dim H \geq 3$*, every $\sigma$-additive state on* $L(H)$ *is of the form* $s(P) = \mathrm{tr}(\rho P)$ *for a unique density operator* $\rho$ *on* $H$*.*

The analogue of Stone duality for OMLs is the McDonald–Bimbó duality [13].

Let $A$ be an OML. A *filter* on $A$ is a non-empty subset $x \subseteq A$ that is upward closed ($a \in x$ and $a \leq b$ implies $b \in x$) and closed under finite meets ($a, b \in x$ implies $a \wedge b \in x$). A filter is *proper* if $0 \notin x$; the unique *improper* filter is $\omega = A$.

For each $a \in A$, the *principal filter* generated by $a$ is $\uparrow a = \{b \in A : a \leq b\}$. Write $F(A)$ for the collection of all filters on $A$, and $P(A) \subseteq F(A)$ for the subcollection of principal filters.

**Definition 2.9** (McDonald–Bimbó [13]). Let $A$ be an OML. The *dual space* of $A$ is the ordered relational topological space

$$S_0(A) = \big(F(A),\ \subseteq,\ \perp_A,\ P(A),\ \mathcal{T}\big),$$

where:

(i) $F(A)$ is the set of all filters on $A$;

(ii) $\subseteq$ is set inclusion;

(iii) $x \perp_A y$ iff there exists $a \in A$ with $a \in x$ and $a^\perp \in y$;

(iv) $P(A)$ is the collection of principal filters;

(v) $\mathcal{T}$ is the topology generated by the subbasis $\{h(a) : a \in A\} \cup \{F(A) \setminus h(a) : a \in A\}$, where $h(a) = \{x \in F(A) : a \in x\}$.

**Theorem 2.10** (McDonald–Bimbó [13]). *$S_0(A)$ is an orthomodular space (compact, totally disconnected). The category of OMLs and homomorphisms is dually equivalent to the category of orthomodular spaces and continuous weak p-morphisms. Moreover, $A \cong \mathrm{CO}(S_0(A))^\dagger$, the OML of clopen $\perp$-stable subsets of $S_0(A)$.*

The critical differences from Stone duality:

| Feature | Boolean (Stone) | OML (McDonald–Bimbó) |
|---|---|---|
| Dual points | Ultrafilters | All filters |
| Distinguished subset | None | $P(A)$ (principal filters) |
| Clopens | Boolean algebra | OML |
| Orthogonality | Trivial | Non-trivial ($\perp_A$) |
| 2-valued homomorphisms | Plentiful (Zorn) | May not exist (KS) |

In Stone duality, the principal filters (i.e., the principal ultrafilters pure($\omega$) for $\omega \in \Omega$) are *invisible*: they play no distinguished role in the dual space. Which ultrafilters are principal depends on the choice of $\Omega$ [12].

In the McDonald–Bimbó duality, $P(A)$ is *part of the structure* of the dual space: the orthomodular frame conditions (Definition 2.9) explicitly reference $P(A)$. This is the algebraic formalisation: for non-distributive algebras, the distinction between realised and unrealised observation-patterns is constrained by the algebra. Distributivity is the condition under which this constraint vanishes.

$S_0(A)$ is compact, so the topological machinery applies. But the algebraic input is weaker. A state $s$ on $A$ defines

$$\mu(h(a)) = s(a)$$

on $\mathrm{CO}(S_0(A))^\dagger$.

In the Boolean case, $\mathrm{Clop}(\mathrm{St}(\mathcal{C}))$ is a Boolean algebra of sets, and $s$ is finitely additive on it. The Carathéodory–Choksi extension machinery applies: compactness of $\mathrm{St}(\mathcal{C})$ gives $\sigma$-subadditivity, and the extension to a Baire measure is automatic.

In the OML case, $\mathrm{CO}(S_0(A))^\dagger$ is an OML of sets — closed under $\cap$ and $^\perp$ but *not* under arbitrary $\cup$. The state $s$ is additive on orthogonal pairs only. Carathéodory does not apply: there is no Boolean algebra of sets on which $s$ is finitely additive.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason's theorem (Theorem 2.8) provides the extension: every state arises from a density operator, giving a $\sigma$-additive measure via the Born rule. For general OMLs, the extension problem is open.

Even when probabilistic descent succeeds, value-definite descent does not. The Bub–Clifton theorem [2] addresses what *can* be salvaged:

**Theorem 2.11** (Bub–Clifton [2])**.** *Given a state $\rho$ on $L(H)$ and a preferred observable $R$, there is a unique maximal sublattice $D(\rho, R) \subseteq L(H)$ such that:*

*(i) $D(\rho, R)$ is a Boolean algebra;*

*(ii) $D(\rho, R)$ contains the spectral projections of $R$;*

*(iii) The restriction of $\rho$ to $D(\rho, R)$ is a mixture of dispersion-free states.*

In the present vocabulary, condition (iii) says that $D(\rho, R)$ is the maximal Boolean subalgebra on which VDR is available — restating Halvorson and Clifton [7] in the EA/PR/VDR framework. It depends on $R$: different preferred observables yield different maximal Boolean subalgebras.

# 3 Distributivity as the commensurability condition

The status of the three positions depends on the algebra:

$$\underbrace{\text{EA}}_{\text{always}} \xleftarrow{\sigma\text{-add.}} \underbrace{\text{PR}}_{\text{Gleason}} \xleftarrow{\text{VDR}} \underbrace{\text{dispersion-free}}_{\text{Zorn (Boolean) / KS-blocked (OML)}}$$

The rightmost arrow is free in the Boolean case and obstructed in the OML case.

**Theorem 3.1** (Commensurability)**.** *Let $A$ be a complemented lattice.*

*(a) If $A$ is distributive (Boolean), then EA, PR, and VDR are mutually commensurable:*

- *EA is unconditional (Stone construction).*
- *PR holds iff the charges are $\sigma$-additive* [12].
- *VDR holds unconditionally (ultrafilters exist).*

*Moreover, the choice of realisation space $\Omega$ is unconstrained by the algebra.*

*(b) If $A$ is non-distributive (OML with $A \cong L(H)$, $\dim H \geq 3$), then:*

- *EA is unconditional.*
- *PR holds (Gleason, Theorem 2.8).*
- *VDR fails globally (Kochen–Specker, Theorem 2.7).*
- *VDR holds locally within each Boolean context (Bub–Clifton, Theorem 2.11).*

*Proof.* Part (a) is Proposition 1.1. For part (b): EA follows from the existence of states on $L(H)$ and the McDonald–Bimbó dual [13]; PR from Gleason's theorem; VDR-failure from Kochen–Specker; local VDR from Bub–Clifton. $\square$

**Corollary 3.2.** *For OMLs of the form $L(H)$ with $\dim H \geq 3$, distributivity is the exact condition under which value-definite realism and empirical adequacy are commensurable.*

*Proof.* If $A$ is distributive, VDR holds (Theorem 3.1(a)). If $A \cong L(H)$ with $\dim H \geq 3$, then $A$ is non-distributive and VDR fails (Theorem 3.1(b)). □

**Remark 3.3.** The dimensional restriction is necessary. For $\dim H = 2$, $L(\mathbb{C}^2)$ is non-distributive but admits dispersion-free states. More generally, horizontal sums of Boolean algebras and certain finite OMLs are non-distributive yet admit 2-valued homomorphisms [10, Chapter 4]. The sharp characterisation for general OMLs is: VDR is available iff $A$ admits a dispersion-free state, which is a property of $A$, not of distributivity alone. Distributivity is sufficient (ultrafilters exist) but not necessary in full generality. For the physically and epistemologically central case $\dim H \geq 3$, it is the effective dividing line.

The underlying structure is a filtration of the state space, whose three transitions require different keys.

**Definition 3.4** (Coherence filtration)**.** Let $A$ be a complemented lattice with a family of charges. The *state space* of $A$ admits a descending filtration:

$$\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{\mathrm{df}}(A),$$

where:

- $\mathcal{S}(A)$ is the set of states on $A$ (contextual coherence);
- $\mathcal{S}_\sigma(A) \subseteq \mathcal{S}(A)$ is the subset of $\sigma$-additive states (probabilistic coherence);
- $\mathcal{S}_{\mathrm{df}}(A) \subseteq \mathcal{S}_\sigma(A)$ is the subset of dispersion-free states (value-definite coherence).

Beneath the filtration sits a local condition: *consistency* requires only that each maximal Boolean subalgebra of $A$ individually admits a compatible state.

Each inclusion can be strict. The three transitions have different mathematical character:

| Transition | Character | Key | Obstruction |
|---|---|---|---|
| Consistency $\to \mathcal{S}(A)$ | Pasting (topological) | Do local sections glue? | Contextuality |
| $\mathcal{S}(A) \to \mathcal{S}_\sigma(A)$ | Regularity (analytic) | Does the state have good limits? | $\sigma$-additivity |
| $\mathcal{S}_\sigma(A) \to \mathcal{S}_{\mathrm{df}}(A)$ | Sharpness (algebraic) | Can probabilities collapse to $\{0,1\}$? | Kochen–Specker |

Distributivity collapses the filtration. For a Boolean algebra:

- *Pasting* is trivial: every local assignment extends to a global state (distributivity guarantees compatibility).
- *Sharpness* is free: ultrafilters exist (Zorn's lemma), so $\mathcal{S}_{\mathrm{df}}(A) \neq \varnothing$ unconditionally.
- The only non-trivial transition is *regularity*: $\sigma$-additivity of the charges. This is the content of [12].

For an OML with $\dim \geq 3$, every transition is non-trivial:

- *Pasting* is non-trivial: contextuality obstructions are possible.
- *Regularity* requires Gleason-type results.
- *Sharpness* is blocked: $\mathcal{S}_{\mathrm{df}}(A) = \varnothing$ (Kochen–Specker).

The filtration is thus a diagnostic: distributivity is the condition that makes it degenerate.

**Remark 3.5** (Relation to contextuality hierarchies)**.** Abramsky and Brandenburger [1] introduced a hierarchy of contextuality — probabilistic, possibilistic, and strong — grading the *strength of the obstruction* to a global section of a presheaf of probability distributions. Our filtration is complementary: it grades *what the algebra can support*, not how badly a global section fails. Their axis measures obstruction strength; ours measures coherence demands.

# 4 Discussion

The standard philosophical debate between scientific realism and constructive empiricism treats the question as uniform: either theories aim at truth or they aim at empirical adequacy. The algebraic analysis above shows that the question is *parametrised* by the observation algebra. For Boolean observations, the mathematics does not prefer one position over another. For non-Boolean observations (dim $\geq 3$), VDR is algebraically impossible; the constructive empiricist position [15] is the generic one, and value-definite realism is the special case — the case in which the observation algebra happens to be distributive.

This reframing retroactively sharpens classical debates. Einstein's realism ("God does not play dice") is the demand for VDR; the Kochen–Specker theorem shows it is algebraically impossible, not merely philosophically contestable. Bohr's complementarity is the non-triviality of the pasting transition. Bell's beables [7] — the observables that *can* have definite values — are the elements of the maximal Boolean subalgebra $D(\rho, R)$ (Theorem 2.11). De Finetti's operationalism is EA in the Boolean case; Pitowsky [14] develops the quantum analogue via Dutch book arguments on $L(H)$.

The analysis is not specific to quantum mechanics. Dzhafarov and Kujala [5] demonstrate orthomodular contextuality in behavioural science; similar structures arise in distributed systems with interfering sensors and experimental designs with mutually exclusive protocols. Wherever observations are incompatible, empirical adequacy entails value-definiteness only if the observation algebra is distributive.

Several directions remain open:

1. *The OML extension problem.* Does every state on a general OML extend to a $\sigma$-additive measure on $S_0(A)$, as in the Boolean case? Gleason's theorem answers this for $L(H)$; the general case is open (§2).

2. *Intermediate algebras.* Between Boolean algebras and arbitrary OMLs lie the $\mathrm{MO}_n$-free OMLs, modular ortholattices, and other subvarieties. Where exactly does VDR fail in this spectrum?

3. *Dynamical structure.* If the observation algebra carries a group action (stationarity, exchangeability), how does symmetry interact with the coherence hierarchy? The Boolean case is partially understood (de Finetti's theorem for exchangeability); the OML case is unexplored.

4. *The topos connection.* The topos approach to quantum theory [9, 4, 8] addresses closely related questions via spectral presheaves rather than the McDonald–Bimbó filter-space duality. In particular, the Bohrification programme of Heunen, Landsman, and Spitters [8] constructs a topos from the poset of Boolean subalgebras — the same contexts that appear in our filtration. A comparison of the two frameworks — and whether the commensurability theorem has a topos-theoretic formulation — is a natural next step.

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